

FRAMEWALA

CONTRIBUTION & DEVELOPMENT STATUS REPORT

Topic	Full-Stack E-Commerce Web Application Development & AI-Assisted Product Customization
Project	FrameWala — Photo Frame & Customised Gift E-Commerce Website
Primary Role	Full-Stack Developer & System Integrator
Project Type	Responsive E-Commerce Website with Photo Frame Personalization
Main Users	Customers and Business Administrator
Live Site	Not yet deployed / deployment to be finalized
Repository	github.com/Varadchavan1129/FrameWala.online
Project Status	Core e-commerce modules implemented; FrameVision prototype under separate development; final integration pending

1. Project Overview

FrameWala is an e-commerce web application for photo frames and customised gift products. The system allows customers to browse products, view details, add items to cart or wishlist, complete checkout, track orders, and submit reviews. A separate FrameVision computer-vision prototype is being developed to detect photo areas in different frame patterns so customer photos can later be placed automatically. The current development focus is on photo frames.

2. Functional Requirements

- Users can browse the home page, categories, active products, and product details.
- Users can search/filter products and view images, price, stock, customization status, and reviews.
- Users can register/login and use guest checkout where applicable.
- Users can add, update, and remove cart items with stock safeguards.
- Users can add/remove wishlist items.
- Users can manage/select saved delivery addresses during checkout.
- Users can place orders through the Razorpay payment flow.
- Users can view order history, track status, and cancel eligible orders.
- Users can submit, edit, and delete their own 1–5 star reviews.
- Admins can manage products, categories, orders, customers, and review moderation.
- Admins can upload multiple product images and manage stock/customization status.
- Customers can upload photos for personalized frame products.
- FrameVision separately detects photo slots/shapes and supports correction/learning workflows.

3. Non-Functional Requirements

- Responsive across desktop, tablet, and mobile.
- Fast, clean, reusable, component-based architecture.
- Protected customer/admin routes and secure authentication.
- Validation for accounts, products, addresses, quantities, reviews, and orders.
- Consistent stock handling across cart, order, and cancellation operations.
- Clear success, error, and confirmation states.
- Scalable for additional products, frame templates, and future customization.
- Clear separation of frontend, backend, services, and database modules.

4. Module-Wise Development Status

Module	Status	Current Development Status
Home / Navigation	Completed	Core customer-facing pages and responsive navigation implemented.
Product Catalog	Completed	Products, categories, search/filtering, stock status implemented.
Product Details	Completed	Details, multiple images, customization status, related products, reviews.
Authentication	Completed	JWT/session handling, protected routes, admin restrictions.

Cart	Completed	Add/update/remove items, quantity safeguards, subtotal.
Wishlist	Completed	Add/remove wishlist functionality.
Checkout / Address	Completed	Checkout forms and saved-address workflow.
Orders / Tracking	Completed	Order history, tracking timeline, cancellation and stock restoration.
Reviews / Ratings	Completed	1–5 star reviews, edit/delete own reviews, admin moderation.
Admin Dashboard	Completed	Metrics and product/category/order/customer management.
Product Management	Completed	CRUD, multiple images, validation, search/filter/sort, stock handling.
Razorpay Payment	To Verify	Gateway selected/integrated; final end-to-end verification pending.
Frame Customization	In Progress	Customer photo upload and customization workflow under development.
FrameVision AI	Prototype / In Progress	Photo-slot detection, shape classification, confidence and correction workflow.
Deployment	To Finalize	Production deployment/domain configuration to be finalized.

5. Assigned Responsibilities

- Define FrameWala requirements, customer flow, admin workflow, and MVP scope.
- Develop React frontend, reusable components, state management, and API services.
- Develop/integrate Node.js + Express backend and MySQL data layer.
- Implement authentication, cart, wishlist, checkout, orders, tracking, reviews, and admin restrictions.
- Implement product/category management, image upload, validation, stock handling, search and filtering.
- Develop the frame personalization workflow for customer photo uploads.
- Develop the separate FrameVision computer-vision prototype.
- Maintain Git/GitHub structure and prepare deployment/integration.

6. Work Completed / Contributed So Far

- Completed the main FrameWala e-commerce architecture using React/Vite/Tailwind, Node.js/Express, and MySQL.
- Implemented authentication, protected routes, product catalog, categories, product details, cart, wishlist, checkout, orders, reviews, and admin modules.
- Implemented stock safeguards and cancellation-based stock restoration.
- Implemented multiple product images and administrator product management.
- Designed the frame personalization architecture while keeping FrameVision separate for testing.
- Developed FrameVision features for photo-slot detection, shape classification, confidence scoring, correction logging, benchmarking, and layout understanding.
- Prepared the future workflow where a frame pattern can be scanned to identify photo areas and support automatic customer-photo placement.

7. Issues / Practical Challenges Addressed

- Keeping authentication and customer/admin permissions consistent.
- Preventing invalid cart quantities and maintaining stock consistency.
- Managing multiple product images and admin-side uploads.
- Designing cart handling so different customized versions can remain separate.
- Supporting frame layouts with different numbers and shapes of photo slots.
- Avoiding manual dimensions for every frame by testing reusable computer-vision detection.
- Keeping FrameVision separate until its detection is sufficiently validated.

8. Pending Work / Improvements

- Complete end-to-end Razorpay verification and production configuration.
- Finalize deployment, domain, environment variables, and production setup.

- Complete customer-facing frame customization UI.
- Validate FrameVision across available frame patterns and improve using correction feedback.
- Integrate FrameVision into FrameWala after standalone testing.
- Add automated testing and complete final documentation, screenshots, diagrams, and demonstration.

9. Current Status of My Contribution

My contribution focuses on full-stack development and system integration of FrameWala, including authentication, product/catalog management, cart, wishlist, checkout, orders, reviews, and administrator functions. In parallel, FrameVision is being developed as a separate computer-vision prototype for frame/photo-slot detection. Core e-commerce modules are implemented, while final customization, production verification, and FrameVision integration remain in progress.

10. Technology Context

Layer	Technology
Frontend	React / Vite / Tailwind CSS
Backend	Node.js + Express
Database	MySQL
API	Axios / REST-style services
Authentication	JWT / Protected Routes
Payment	Razorpay
State Management	React Context API
AI/CV Prototype	Python + FastAPI + OpenCV + SAM-based components
Version Control	Git + GitHub
Currency	INR

11. Team Coordination

- Keep frontend, backend, database, payment, and customization modules aligned.
- Use Git/GitHub for collaborative development and controlled integration.
- Keep FrameVision separate during the testing phase.
- Clearly distinguish implemented, in-progress, and planned features.
- Use the module status table for demonstrations and progress updates.

12. Team Members & Roles

Vedanti Pawar (Member 1)

Role: UI/UX Designer

- Designed the overall user interface structure and visual layout of FrameWala.
- Worked on page flow, navigation, product presentation, and responsive screen layouts.
- Helped maintain a consistent, simple, and user-friendly design across customer and admin pages.
- Prepared UI ideas and design references for the frame customization experience.

Aparna Jadhav (Member 2)

Role: Frontend Developer

- Developed customer-facing frontend pages and reusable React components.
- Implemented product listing, product details, cart, wishlist, checkout, and related frontend interactions.
- Connected frontend screens with application services/API calls where required.
- Worked on responsive layouts and frontend validation/error states.

Pankaj Gode (Member 3)

Role: Requirements Gathering & Client Communication

- Gathered project requirements and discussed business needs with the client.
- Collected information about products, customers, customization requirements, and expected website features.
- Helped convert client requirements into functional modules and development tasks.
- Supported requirement clarification, feedback collection, and coordination between the client and development team.

Varad Chavan (Member 4)

Role: Integration & Application Logic

- Worked on integration of frontend, backend, database, authentication, and major application modules.

- Implemented and coordinated core application logic for cart, wishlist, checkout, orders, reviews, and admin workflows.
- Worked on FrameWala–FrameVision integration planning and the frame customization workflow.
- Handled debugging, feature integration, validation, and overall system coordination.
- Maintained Git/GitHub project integration and helped ensure modules work together correctly.

Rohit Hegadmal (Member 5)

Role: Database & Backend Developer

- Worked on backend development using Node.js and Express.
- Designed and managed MySQL database structures for users, products, categories, carts, orders, reviews, and related data.
- Developed backend APIs and database operations required by the frontend.
- Worked on data validation, CRUD operations, authentication-related backend logic, and database consistency.

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